PHYS 143 – Mechanics and Thermal Physics

Week 4: Work and Kinetic Energy – Tutorial Sheet

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Use the Master Procedure: identify, mark the two states, draw the free-body diagram, find the work of each force with its sign, predict, select the method, execute $W_{\text{tot}} = \Delta K$, evaluate. Use $g = 9.80\,\text{m/s}^2$. Problems A1 and A2 are solved in full, one step per line; final answers to the others are at the end. Problems B9, B10, B12, B13, B15 and B16 were moved from the lecture.

A. Worked examples

A1. A loaded handcart on level ground

A $15.0\,\text{kg}$ loaded handcart is pulled $10.0\,\text{m}$ along level ground by a rope with tension $60\,\text{N}$, 20° above the horizontal. The coefficient of kinetic friction is 0.10. Find the work done by each force, the total work, and the final speed if it starts from rest.

Predict. The tension does positive work and friction negative work, so $W_{\text{tot}} > 0$ and the cart speeds up.

IDENTIFY. Given: $m = 15.0\,\text{kg}$, $T = 60\,\text{N}$ at 20°, $s = 10.0\,\text{m}$, $\mu_k = 0.10$, $v_1 = 0$. Find: each W, W_{tot}, v_2.

SET UP. Forces: $\vec{\boldsymbol{T}}$, $\vec{\boldsymbol{n}}$, $\vec{\boldsymbol{w}}$, $\vec{\boldsymbol{f}}_k$. Works from $W = Fs\cos\phi$; total from $W_{\text{tot}} = \sum W_i = (\sum \vec{\boldsymbol{F}}) \cdot \vec{\boldsymbol{s}}$; speed from $W_{\text{tot}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$.

EXECUTE.

Normal force: $n = mg - T\sin 20^\circ = 147.0\,\text{N} - 20.5\,\text{N} = 126.5\,\text{N}$

Friction: $f_k = \mu_k n = 0.10(126.5\,\text{N}) = 12.65\,\text{N}$

Tension: $W_T = Ts\cos 20^\circ = (60\,\text{N})(10.0\,\text{m})(0.9397) = +564\,\text{J}$

Friction: $W_f = -f_k s = -(12.65\,\text{N})(10.0\,\text{m}) = -127\,\text{J}$; $W_w = W_n = 0$

Total: $W_{\text{tot}} = 564\,\text{J} - 127\,\text{J} = \boxed{437\,\text{J}}$

Check (Method II): $(60\cos 20^\circ - 12.65)(10.0) = (56.4 - 12.65)(10.0)\,\text{J} = 437\,\text{J}$

Speed: $v_2 = \sqrt{2W_{\text{tot}}/m} = \sqrt{2(437\,\text{J})/15.0\,\text{kg}} = \boxed{7.6\,\text{m/s}}$

EVALUATE. $W_{\text{tot}} > 0$ and the cart speeds up, as predicted. The two methods agree. Units: J and m/s.

A2. A spring-launched ball

A $0.300\,\text{kg}$ ball is pushed against a horizontal spring ($k = 500\,\text{N/m}$) in a laboratory launcher, compressing it $0.20\,\text{m}$, and released on a smooth surface. Find its speed when it leaves the spring.

Predict. Smooth surface, so all the spring's work becomes kinetic energy. Of order $10\,\text{J}$, hence several m/s.

IDENTIFY. Given: $m = 0.300\,\text{kg}$, $k = 500\,\text{N/m}$, $x_1 = 0.20\,\text{m}$, $x_2 = 0$, $v_1 = 0$. Find: v_2.

SET UP. The spring force varies, so use the work by the spring, $W_{\text{by}} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2$, and $W_{\text{tot}} =$

$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$. Gravity and the normal force are perpendicular to the motion.

EXECUTE.

Spring: $W_{\text{by}} = \frac{1}{2}(500\,\text{N/m})(0.20\,\text{m})^2 - 0 = 10.0\,\text{J}$

Theorem: $10.0\,\text{J} = \frac{1}{2}(0.300\,\text{kg})v_2^2$

Solve: $v_2 = \sqrt{2(10.0\,\text{J})/0.300\,\text{kg}} = \boxed{8.2\,\text{m/s}}$

EVALUATE. Several m/s, as predicted. Units: $\sqrt{\text{J/kg}} = \text{m/s}$.

B. Practice

B1. A ramp with friction

A $10.0\,\text{kg}$ crate of mangoes slides $4.0\,\text{m}$ down a 30° ramp with $\mu_k = 0.25$, starting from rest. Find the work done by gravity, by friction and by the normal force, and the final speed.

B2. Stopping a tipper truck

An $8000\,\text{kg}$ tipper truck at $20\,\text{m/s}$ is stopped by a constant braking force of $32\,\text{kN}$. Find the stopping distance. How far would it need at $40\,\text{m/s}$?

B3. A varying force

The force on a particle is $F_x = 10x - 2x^2$ (SI units). Find the work done from $x = 0$ to $x = 3.0\,\text{m}$.

B4. A spring and friction

A $1.0\,\text{kg}$ block is pressed against a spring ($k = 200\,\text{N/m}$) compressed $0.30\,\text{m}$ and released on a rough floor ($\mu_k = 0.30$). Find its speed as it leaves the spring and the total distance it slides.

B5. Power of a hoist

A motor on a building site raises an $800\,\text{kg}$ load at a constant $2.0\,\text{m/s}$. What is the power developed (neglecting friction)? How much energy does it supply in one minute?

B6. Challenge: the method is not given

A $60\,\text{kg}$ boy slides down a $25\,\text{m}$ long grassy slope inclined 20° to the horizontal with $\mu_k = 0.10$, and then continues along a level concrete patch with $\mu_k = 0.20$ until he stops. How far does he slide on the level patch?

B7. Letting go of the swing

The $25.0\,\text{kg}$ child of Example 9 sits on a swing with chains $3.30\,\text{m}$ long. She is released from rest with the chains 35° from the vertical. Find her speed at the lowest point. Which forces do work?

B8. Work from components

A force $\vec{\boldsymbol{F}} = (40\,\mathrm{N})\,\hat{\boldsymbol{\imath}} + (30\,\mathrm{N})\,\hat{\boldsymbol{\jmath}}$ acts on an object while its displacement is $\vec{\boldsymbol{s}} = (5.0\,\mathrm{m})\,\hat{\boldsymbol{\imath}} - (2.0\,\mathrm{m})\,\hat{\boldsymbol{\jmath}}$. Find the work done, and the angle between $\vec{\boldsymbol{F}}$ and $\vec{\boldsymbol{s}}$.

B9. Units in context (moved from the lecture)

(a) A $4.0\,\mathrm{g}$ bullet moving at $300\,\mathrm{m/s}$ stops after penetrating $12\,\mathrm{cm}$ of wood. Find the average stopping force. (b) A $3.0\,\mathrm{kW}$ motor runs for $2.0\,\mathrm{h}$: how much energy does it use, in joules and in kilowatt-hours? (c) An electric iron rated $1.2\,\mathrm{kW}$ is used for $3.0\,\mathrm{h}$ a week for 4 weeks. How many kilowatt-hours is that, as billed by ECG?

B10. Work by tension on connected masses (moved from the lecture)

Blocks of $3.0\,\mathrm{kg}$ and $5.0\,\mathrm{kg}$ hang from a light rope over a frictionless, massless pulley and are released from rest. The heavier block descends $2.0\,\mathrm{m}$. Find the work done by the tension on each block, the net work on each block, and the total work done by the rope. Why does the rope do no net work?

B11. A pile driver

A $500\,\mathrm{kg}$ hammerhead is released from rest $2.0\,\mathrm{m}$ above a pile and drives it $0.10\,\mathrm{m}$ into the ground. Find the hammer's speed at impact and the average resisting force the pile exerts on the hammer while it is driven in.

B13. The other side of the spring

In Example 8(b) of the lecture ($m = 0.200\,\mathrm{kg}$, $k = 40.0\,\mathrm{N/m}$, released from rest $0.15\,\mathrm{m}$ from natural length, friction $0.50\,\mathrm{N}$) the trolley passes $x = 0$ and compresses the spring. How far, d, does it go before it stops for an instant?

B14. Power of an aircraft engine

A jet of mass $4.0 \times 10^4\,\mathrm{kg}$ flies at a constant $200\,\mathrm{m/s}$ at constant altitude. Each of its two engines gives a forward thrust of $75\,\mathrm{kN}$. What power does each engine develop, in megawatts? What is the total resistive force on the aircraft?

B15. Two stair climbers (moved from the lecture)

A $60\,\mathrm{kg}$ student climbs $4.0\,\mathrm{m}$ of stairs in $5.0\,\mathrm{s}$. (a) Find the average power. (b) An $80\,\mathrm{kg}$ student develops the same power. How long does the same climb take?

B16. A tractor and a crate of yams (moved from the lecture)

A tractor drags a $50.0\,\mathrm{kg}$ crate of yams $8.0\,\mathrm{m}$ across level ground. The cable force is $500\,\mathrm{N}$, 30° above the horizontal, and $\mu_k = 0.20$. Find the work done by each force and the total work, by both methods.

B17. Work by a third-law pair

(a) A weightlifter lowers a $15.0\,\mathrm{kg}$ barbell $0.80\,\mathrm{m}$ at constant speed. Find the work her hands do on the barbell and the work the barbell does on her hands. (b) A $2.0\,\mathrm{kg}$ block slides $3.0\,\mathrm{m}$ along a fixed table with $\mu_k = 0.30$. Find the work of each friction force of the pair and their sum. Explain the difference between (a) and (b).

B18. Iceboats

Two iceboats of masses $200\,\mathrm{kg}$ and $400\,\mathrm{kg}$ race over $50\,\mathrm{m}$ of frictionless ice. The wind exerts a constant $100\,\mathrm{N}$ on each. Find the kinetic energy and the speed of each at the finish, starting from rest.

B12. Friction that does positive work (moved from the lecture)

A $20\,\mathrm{kg}$ box of yams rests on the bed of a pickup truck. The truck accelerates at $2.0\,\mathrm{m/s^2}$ over $50\,\mathrm{m}$ and the box does not slide. Find the work done on the box by static friction and compare it with the box's change in kinetic energy.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80\,\mathrm{m/s^2}$.

E1. A trader pulls a crate of tomatoes $8.0\,\mathrm{m}$ along a level market floor with a rope inclined at 35° above the horizontal and a constant tension of $60\,\mathrm{N}$. The work done by the tension is

(A) $49.1\,\mathrm{J}$ **(B)** $275\,\mathrm{J}$ **(C)** $393\,\mathrm{J}$ **(D)** $480\,\mathrm{J}$ **(E)** $786\,\mathrm{J}$

E2. A force $F_x = 6x^2 - 4x$ (F in N, x in m) acts on a particle that moves from $x = 0$ to $x = 2.0\,\mathrm{m}$. The work done by the force is

(A) $4.00\,\mathrm{J}$ **(B)** $8.00\,\mathrm{J}$ **(C)** $16.0\,\mathrm{J}$ **(D)** $20.0\,\mathrm{J}$ **(E)** $32.0\,\mathrm{J}$

E3. The net work needed to increase the speed of a $1200\,\mathrm{kg}$ car from $15.0\,\mathrm{m/s}$ to $25.0\,\mathrm{m/s}$ is

(A) $1.20 \times 10^4\,\mathrm{J}$

(B) $6.00 \times 10^4\,\mathrm{J}$

(C) $1.35 \times 10^5\,\mathrm{J}$

(D) $2.40 \times 10^5\,\mathrm{J}$

(E) $3.75 \times 10^5\,\mathrm{J}$

Items E4 and E5 share this information: A $900\,\mathrm{kg}$ car travelling at $20.0\,\mathrm{m/s}$ is brought to rest by a constant braking force over a distance of $40.0\,\mathrm{m}$.

E4. The magnitude of the braking force is

(A) $450\,\mathrm{N}$ **(B)** $2250\,\mathrm{N}$ **(C)** $4500\,\mathrm{N}$ **(D)** $8820\,\mathrm{N}$ **(E)** $9000\,\mathrm{N}$

E5. The same car with the same braking force starts at twice the speed. The stopping distance is

(A) $20\,\mathrm{m}$ **(B)** $80\,\mathrm{m}$ **(C)** $120\,\mathrm{m}$ **(D)** $160\,\mathrm{m}$ **(E)** $320\,\mathrm{m}$

E6. A spring of force constant $400\,\mathrm{N/m}$ is slowly stretched from an extension of $0.050\,\mathrm{m}$ to $0.150\,\mathrm{m}$. The work done by the stretching force is

(A) $4.0\,\mathrm{J}$ **(B)** $4.5\,\mathrm{J}$ **(C)** $8.0\,\mathrm{J}$ **(D)** $9.0\,\mathrm{J}$ **(E)** $40.0\,\mathrm{J}$

E7. Which of the following statements is/are true?

I. A force that is always perpendicular to the displacement does no work.

II. The total work done by the centripetal force during one full revolution of uniform circular motion is zero.

III. Friction can never do positive work.

(A) I only **(B)** I and II only **(C)** I and III only **(D)** II and III only **(E)** I, II and III

Items E8 and E9 share this information: Blocks of mass $3.0\,\mathrm{kg}$ and $5.0\,\mathrm{kg}$ hang from the ends of a light rope over a frictionless, massless pulley and are released from rest. The heavier block descends $2.0\,\mathrm{m}$.

E8. The work done by the rope tension on the $3.0\,\mathrm{kg}$ block (which rises) is

(A) $-73.5\,\mathrm{J}$ **(B)** $0.0\,\mathrm{J}$ **(C)** $58.8\,\mathrm{J}$ **(D)** $73.5\,\mathrm{J}$ **(E)** $98.0\,\mathrm{J}$

E9. The net work done on the $5.0\,\mathrm{kg}$ block as it descends is

(A) $-24.5\,\mathrm{J}$ **(B)** $49.0\,\mathrm{J}$ **(C)** $98.0\,\mathrm{J}$ **(D)** $171.5\,\mathrm{J}$ **(E)** $24.5\,\mathrm{J}$

E10. A $4.0\,\mathrm{g}$ bullet travelling at $300\,\mathrm{m/s}$ stops after penetrating $12\,\mathrm{cm}$ into a wooden block. The average stopping force is

(A) $1.50\,\mathrm{N}$

(B) $15.0\,\mathrm{N}$

(C) $1500\,\mathrm{N}$

(D) $1.50 \times 10^5\,\mathrm{N}$

(E) $1.50 \times 10^6\,\mathrm{N}$

Answers

B1 $W_g = +196\,\text{J}$, $W_f = -85\,\text{J}$, $W_n = 0$; $v = 4.7\,\text{m/s}$
B2 $50\,\text{m}$; $200\,\text{m}$ (four times)
B3 $27\,\text{J}$
B4 $v = 4.0\,\text{m/s}$; $D = 3.1\,\text{m}$
B5 $P = mgv = 15.7\,\text{kW}$ ($\approx 21\,\text{hp}$); $9.4 \times 10^5\,\text{J}$
B6 kinetic energy at the bottom $3.6 \times 10^3\,\text{J}$ ($v = 11\,\text{m/s}$); distance on the level patch $31\,\text{m}$
B7 $v = \sqrt{2gh} = 3.4\,\text{m/s}$ with $h = L(1 - \cos 35^\circ) = 0.597\,\text{m}$; gravity does $+mgh$, the tension does no work
B8 $W = (40)(5.0) + (30)(-2.0) = 140\,\text{J}$; $F = 50\,\text{N}$, $s = 5.39\,\text{m}$, $\phi = 58.7^\circ$
B9 (a) $1.5 \times 10^3\,\text{N}$ (leaving d in cm gives $30\,\text{N}$); (b) $2.2 \times 10^7\,\text{J} = 6.0\,\text{kWh}$; (c) $14.4\,\text{kWh} = 5.2 \times 10^7\,\text{J}$
B10 $T = 36.75\,\text{N}$: $+73.5\,\text{J}$ on the $3.0\,\text{kg}$ block, $-73.5\,\text{J}$ on the $5.0\,\text{kg}$ block; net work $14.7\,\text{J}$ and $24.5\,\text{J}$; total $39.2\,\text{J} = (m_2 - m_1)gs$; the rope only transfers energy
B11 $v = \sqrt{2gh} = 6.3\,\text{m/s}$; $R = mg(h + d)/d = 1.0 \times 10^5\,\text{N}$
B12 $f_s = 40\,\text{N}$ forward, $W = +2000\,\text{J} = \Delta K$
B13 $20d^2 + 0.50d - 0.375 = 0$, so $d = 0.125\,\text{m}$
B15 $P = 470\,\text{W}$; $t = 6.7\,\text{s}$
B16 $n = 240\,\text{N}$, $f_k = 48\,\text{N}$; $W_T = +3.46 \times 10^3\,\text{J}$, $W_f = -384\,\text{J}$, $W_w = W_n = 0$; $W_{\text{tot}} = 3.08 \times 10^3\,\text{J}$ (net force $385\,\text{N}$ times $8.0\,\text{m}$)
B17 (a) $F = 147\,\text{N}$: $-118\,\text{J}$ on the barbell, $+118\,\text{J}$ on the hands, sum zero (same displacement); (b) $f = 5.88\,\text{N}$: $-17.6\,\text{J}$ on the block, 0 on the table (it does not move), sum $-17.6\,\text{J}$
B18 $K = Fs = 5000\,\text{J}$ for both; $v = 7.1\,\text{m/s}$ and $5.0\,\text{m/s}$
B14 $P = Fv = (75\,000)(200) = 1.5 \times 10^7\,\text{W} = 15\,\text{MW}$ per engine; constant velocity, so the resistive force equals the total thrust, $150\,\text{kN}$

Exam-style practice: answer key and brief solutions

Key: E1 C E2 B E3 D E4 C E5 D E6 A E7 B E8 D E9 E E10 C

E1 (C). $W = Fs\cos\phi$ with $\phi = 35^\circ$, the angle between the force and the displacement: $W = (60)(8.0)\cos 35^\circ = 393\,\text{J}$. Only the component along the motion counts, so $W < Fs = 480\,\text{J}$.

E2 (B). $W = \int_0^{2.0}(6x^2 - 4x)\,\text{d}x = \left[2x^3 - 2x^2\right]_0^{2.0} = 16 - 8 = 8.0\,\text{J}$. The final force $F(2.0) = 16\,\text{N}$ times the distance would give $32\,\text{J}$, but the force is not constant: it is zero at $x = 0.67\,\text{m}$ and negative before that, so integrate.

E3 (D). $W_{\text{net}} = \Delta K = \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2}(1200)(625 - 225) = 2.40 \times 10^5\,\text{J}$. The squares are subtracted, not the speeds: using $(v_2 - v_1)^2$ gives $6.0 \times 10^4\,\text{J}$.

E4 (C). Work–energy: $-Fd = 0 - \frac{1}{2}mv^2$, so $F = \dfrac{mv^2}{2d} = \dfrac{(900)(400)}{2(40.0)} = 4500\,\text{N}$. The braking force does negative work, so $W = -Fd = -1.8 \times 10^5\,\text{J} = \Delta K$.

E5 (D). $d = \dfrac{mv^2}{2F} \propto v^2$. Doubling v multiplies the distance by 4: $d = 160\,\text{m}$. The work needed is four times larger because $K \propto v^2$.

E6 (A). $W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2}(400)(0.0225 - 0.0025) = 4.0\,\text{J}$. The squares are subtracted: the work from the natural length to $0.150\,\text{m}$ alone is $4.5\,\text{J}$, which includes the first $0.050\,\text{m}$.

E7 (B). I is true: $W = Fs\cos 90^\circ = 0$. II is true: the centripetal force is always perpendicular to the velocity, so the speed and K do not change. III is false: static friction does positive work on a box carried by an accelerating lorry or conveyor belt. Answer: I and II only.

E8 (D). For the pair $a = \dfrac{m_2 - m_1}{m_1 + m_2} g = 2.45\,\mathrm{m/s^2}$ and the tension is $T = \dfrac{2m_1 m_2}{m_1 + m_2} g = 36.75\,\mathrm{N}$. The tension is upward, along the displacement of the rising block: $W_T = T\,s = (36.75)(2.0) = 73.5\,\mathrm{J}$, positive. (On the descending block the same tension does $-73.5\,\mathrm{J}$, so the rope does no net work.)

E9 (E). Net work $= W_w + W_T = m_2 g h - T s = (5.0)(9.80)(2.0) - (36.75)(2.0) = 98.0 - 73.5 = 24.5\,\mathrm{J}$, which equals its gain in kinetic energy $\tfrac{1}{2} m_2 v^2$ with $v^2 = 2as = 9.80$. Gravity alone does $98.0\,\mathrm{J}$, but the tension takes some of it away.

E10 (C). Convert before substituting: $m = 4.0 \times 10^{-3}\,\mathrm{kg}$, $d = 0.12\,\mathrm{m}$. Then $F = \dfrac{mv^2}{2d} = \dfrac{(4.0 \times 10^{-3})(300)^2}{2(0.12)} = 1500\,\mathrm{N}$. Leaving d in cm gives $30\,\mathrm{N}$, and leaving m in grams gives $1.5 \times 10^6\,\mathrm{N}$: check the magnitude against a bullet that stops in a few centimetres.